Bibliometric Co-occurrence Network — Sarcoptes scabiei Diagnostics

Built from diagnostic accuracy study data

Overview

This network visualises co-occurrence relationships in the diagnostic accuracy literature on scabies (*Sarcoptes scabiei*). Each edge represents a study that links S. scabiei to a host, a diagnostic method, a sample type, or an antigen type. The thicker the edge, the more studies support that link.

```
library(tidyverse)
library(janitor)

raw <- read_csv("../data/Diagnostic_Accuracy_Table.csv", show_col_types = FALSE)

# Clean data: remove non-ASCII, filter valid rows
d <- raw |>
  clean_names() |>
  filter(!is.na(tp)) |>
  mutate(across(where(is.character), ~ iconv(., from = "latin1", to = "ASCII//TRANSLIT"))) |>
  mutate(across(where(is.character), ~ gsub("[^ -~]", "", .)))

# --- Categorise variables for the network ---

d <- d |> mutate(
  host_group = case_when(
    tolower(study_on) == "human" ~ "Humans",
    TRUE ~ "Animals"
  ),
  method_group = case_when(
    method %in% c("qPCR", "Qpcr", "nqPCR", "nPCR", "cPCR") ~ "qPCR / cPCR",
    method == "LAMP" ~ "LAMP",
```

```r
    method == "nRT-qPCR" ~ "RT-qPCR",
    grepl("ELISA|DELFIA", method, ignore.case = TRUE) ~ "ELISA"
  ),
  sample_group = case_when(
    tolower(sample_type) == "serum" ~ "Serum",
    grepl("scrap", tolower(sample_type)) ~ "Skin scraping",
    grepl("swab", tolower(sample_type)) ~ "Swab",
    TRUE ~ sample_type
  ),
  antigen_group = case_when(
    grepl("wma", tolower(antigen_type)) ~ "Whole antigen",
    grepl("recombinant", tolower(antigen_type)) ~ "Recombinant",
    tolower(antigen_type) %in% c("mtdna", "nuclear genes", "microsatellite marker") ~ "DNA ta
  )
)

# --- Build co-occurrence edge weights ---

# Edge: Sarcoptes   host
host_edges <- d |> count(host_group, name = "weight") |> mutate(to = host_group, from = "Sar
# Edge: Sarcoptes   method
method_edges <- d |> filter(!is.na(method_group)) |> count(method_group, name = "weight") |>
# Edge: Sarcoptes   sample type
sample_edges <- d |> count(sample_group, name = "weight") |> mutate(to = sample_group, from =
# Edge: Sarcoptes   antigen type
antigen_edges <- d |> filter(!is.na(antigen_group)) |> count(antigen_group, name = "weight")

# Edge: Serum   ELISA (all 40 ELISA studies used serum, but let's count)
serum_elisa <- d |> filter(method_group == "ELISA", sample_group == "Serum") |> nrow()

all_edges <- bind_rows(host_edges, method_edges, sample_edges, antigen_edges)
all_edges <- all_edges |> filter(!is.na(to))
```

Co-occurrence network

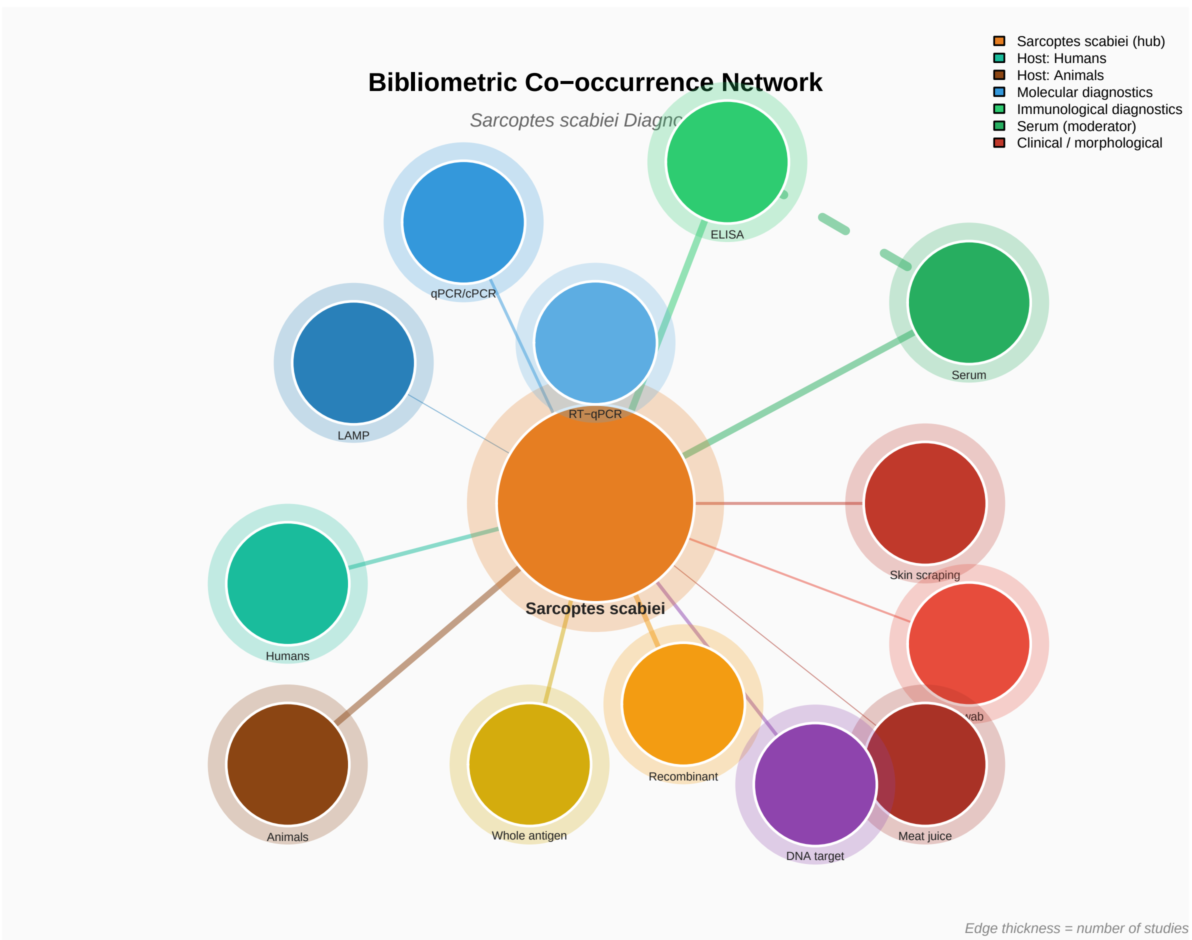
```r
# ---- Layout: assign (x, y) positions for each node ----

# Outer ring: node groups with their positions
```

```r
all_nodes <- c("Sarcoptes scabiei",
  "Humans", "Animals",
  "ELISA", "qPCR / cPCR", "LAMP", "RT-qPCR",
  "Serum", "Skin scraping", "Swab", "meat-juice",
  "Whole antigen", "Recombinant", "DNA target")

# Positions - compact, hub at centre
node_pos <- list(
  "Sarcoptes scabiei" = c(0, 0),
  "Humans" = c(-1.4, -0.4),
  "Animals" = c(-1.4, -1.3),
  "ELISA" = c(0.6, 1.7),
  "qPCR / cPCR" = c(-0.6, 1.4),
  "LAMP" = c(-1.1, 0.7),
  "RT-qPCR" = c(0.0, 0.8),
  "Serum" = c(1.7, 1.0),
  "Skin scraping" = c(1.5, 0.0),
  "Swab" = c(1.7, -0.7),
  "meat-juice" = c(1.5, -1.3),
  "Whole antigen" = c(-0.3, -1.3),
  "Recombinant" = c(0.4, -1.0),
  "DNA target" = c(1.0, -1.4)
)

# Colours for groups
group_colors <- c(
  "Sarcoptes scabiei" = "#E67E22",
  "Humans" = "#1ABC9C",
  "Animals" = "#8B4513",
  "ELISA" = "#2ECC71",
  "qPCR / cPCR" = "#3498DB",
  "LAMP" = "#2980B9",
  "RT-qPCR" = "#5DADE2",
  "Serum" = "#27AE60",
  "Skin scraping" = "#C0392B",
  "Swab" = "#E74C3C",
  "meat-juice" = "#A93226",
  "Whole antigen" = "#D4AC0D",
  "Recombinant" = "#F39C12",
  "DNA target" = "#8E44AD"
)
```

```r
# Labels for groups (display names)
display_name <- c(
  "Sarcoptes scabiei" = "Sarcoptes scabiei",
  "Humans" = "Humans",
  "Animals" = "Animals",
  "ELISA" = "ELISA",
  "qPCR / cPCR" = "qPCR/cPCR",
  "LAMP" = "LAMP",
  "RT-qPCR" = "RT-qPCR",
  "Serum" = "Serum",
  "Skin scraping" = "Skin scraping",
  "Swab" = "Swab",
  "meat-juice" = "Meat juice",
  "Whole antigen" = "Whole antigen",
  "Recombinant" = "Recombinant",
  "DNA target" = "DNA target"
)

# ---- Plot ----

# Set up blank canvas
par(mar = c(0, 0, 0, 0), bg = "#FAFAFA")
plot(NA, xlim = c(-2.5, 2.5), ylim = c(-2.0, 2.3),
     xlab = "", ylab = "", xaxt = "n", yaxt = "n", bty = "n")

# Title
text(0, 2.1, "Bibliometric Co-occurrence Network", cex = 0.9, font = 2, xpd = TRUE)
text(0, 1.9, "Sarcoptes scabiei Diagnostics", cex = 0.65, font = 3, col = "#666666", xpd = T

# Draw edges: Sarcoptes   each other node
# Edge width proportional to sqrt(weight) for visual clarity
for (i in seq_len(nrow(all_edges))) {
  e <- all_edges[i, ]
  x0 <- node_pos[[e$from]][1]
  y0 <- node_pos[[e$from]][2]
  x1 <- node_pos[[e$to]][1]
  y1 <- node_pos[[e$to]][2]

  if (!is.null(x0) && !is.null(x1)) {
    w <- sqrt(e$weight) * 0.6
    segments(x0, y0, x1, y1,
             col = adjustcolor(group_colors[e$to], alpha.f = 0.5),
```

```
             lwd = w)
  }
}

# Special edge: Serum → ELISA (moderator relationship)
x0 <- node_pos[["Serum"]][1]
y0 <- node_pos[["Serum"]][2]
x1 <- node_pos[["ELISA"]][1]
y1 <- node_pos[["ELISA"]][2]
segments(x0, y0, x1, y1,
         col = adjustcolor("#27AE60", alpha.f = 0.5), lwd = sqrt(serum_elisa) * 0.8, lty = 2)

# Draw nodes - scaled down
for (node in all_nodes) {
  pos <- node_pos[[node]]
  if (!is.null(pos)) {
    col <- group_colors[node]
    # Base circle size: hub bigger
    r <- ifelse(node == "Sarcoptes scabiei", 0.45, 0.28)
    # Outer glow
    symbols(pos[1], pos[2], circles = r * 1.3,
            inches = FALSE, add = TRUE,
            bg = adjustcolor(col, alpha.f = 0.25),
            fg = NA)
    # Main node
    symbols(pos[1], pos[2], circles = r,
            inches = FALSE, add = TRUE,
            bg = col,
            fg = "white", lwd = 1.5)
    # Label
    lbl <- display_name[node]
    cex_lbl <- ifelse(node == "Sarcoptes scabiei", 0.55, 0.45)
    font_lbl <- ifelse(node == "Sarcoptes scabiei", 2, 1)
    text(pos[1], pos[2] - r - 0.08,
         lbl, cex = cex_lbl, font = font_lbl,
         col = "#222222", xpd = TRUE)
  }
}

# Legend - smaller
legend("topright",
  legend = c(
```

```
    "Sarcoptes scabiei (hub)",
    "Host: Humans", "Host: Animals",
    "Molecular diagnostics", "Immunological diagnostics",
    "Serum (moderator)", "Clinical / morphological"
  ),
  fill = c("#E67E22", "#1ABC9C", "#8B4513",
           "#3498DB", "#2ECC71",
           "#27AE60", "#C0392B"),
  cex = 0.5, bty = "n", title = "", title.adj = 0,
  xpd = TRUE
)

# Note
mtext("Edge thickness = number of studies", side = 1, line = -1,
      cex = 0.5, col = "#888888", font = 3, adj = 1)
```



Edge weights table

```r
all_edges |>
  select(from, to, weight) |>
  arrange(desc(weight)) |>
  knitr::kable(caption = "Co-occurrence edge weights (number of studies)")
```

Table 1: Co-occurrence edge weights (number of studies)

from	to	weight
Sarcoptes scabiei	ELISA	42
Sarcoptes scabiei	Serum	40
Sarcoptes scabiei	Animals	37
Sarcoptes scabiei	Recombinant	23
Sarcoptes scabiei	Humans	16
Sarcoptes scabiei	Whole antigen	16
Sarcoptes scabiei	DNA target	11
Sarcoptes scabiei	qPCR / cPCR	8
Sarcoptes scabiei	Skin scraping	8
Sarcoptes scabiei	Swab	4
Sarcoptes scabiei	RT-qPCR	2
Sarcoptes scabiei	LAMP	1
Sarcoptes scabiei	meat-juice	1

```r
cat("ELISA studies using Serum:", serum_elisa, "out of",
    sum(d$method_group == "ELISA", na.rm = TRUE), "total ELISA studies")
```

```
ELISA studies using Serum: 40 out of 42 total ELISA studies
```

Interpretation

- **Sarcoptes scabiei** (orange, centre) connects to all other nodes.
- **Hosts:** 16 studies on humans (teal), 37 on animals (brown — pigs, rabbits, dogs, foxes, sheep, chamois, bears).

- **Molecular diagnostics** (blues): qPCR/cPCR (8 studies), LAMP (1), RT-qPCR (2).
- **Immunological diagnostics** (green): ELISA dominates with 42 studies. Nearly all ELISA studies used **serum** (dashed green edge) — serum acts as a *moderator* variable between the diagnostic method and the pathogen.
- **Clinical/morphological** (reds): Skin scrapings and swabs are the sample collection methods used across both diagnostic approaches.
- **Antigen types** (golds/purple): Whole antigen (16), recombinant (23), DNA target (11).
- Edge thickness reflects the number of linking studies — more evidence behind that connection.